

Block-encoding K matrices A_k of fixed size $256 \times 256 = 2^8 \times 2^8$

$K = k^3$ for k in $[2, 3, 4, 5, 6]$ (so K in $[8, 27, 64, 125, 216]$); phase_bitsize $b = 32$.

Approach 1 (SVD): $A_k = U_k \Sigma_k V_k$.

Cost = $2 * \text{BlockUnitaryInterferometerSynthesisQROAM}(K, 2^n)$
+ $\text{QROAMClean}((K, 2^n), b) + \text{ctrl-AddIntoPhaseGrad} + \text{QROAMCleanAdjoint}$.

Approach 2 (direct dilation): A_k completed to W_k of size $2^{(n+1)} \times 2^{(n+1)}$.

Cost = $\text{BlockUnitaryInterferometerSynthesisQROAM}(K, 2^{(n+1)})$.

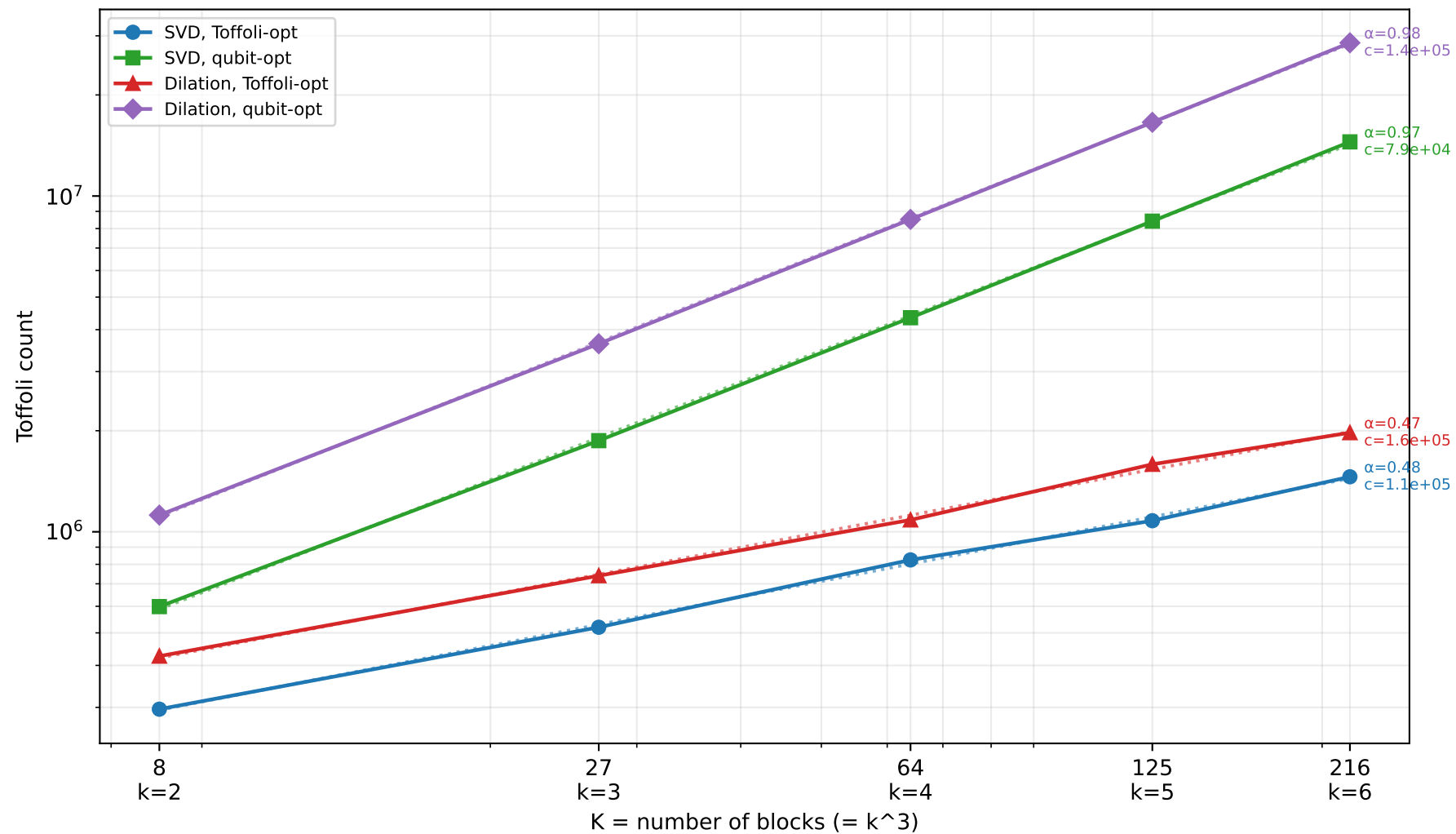
Each cost is optimized over its `log_block_size` sweep.

Two optimization modes per approach: Toffoli-optimal and qubit-optimal.

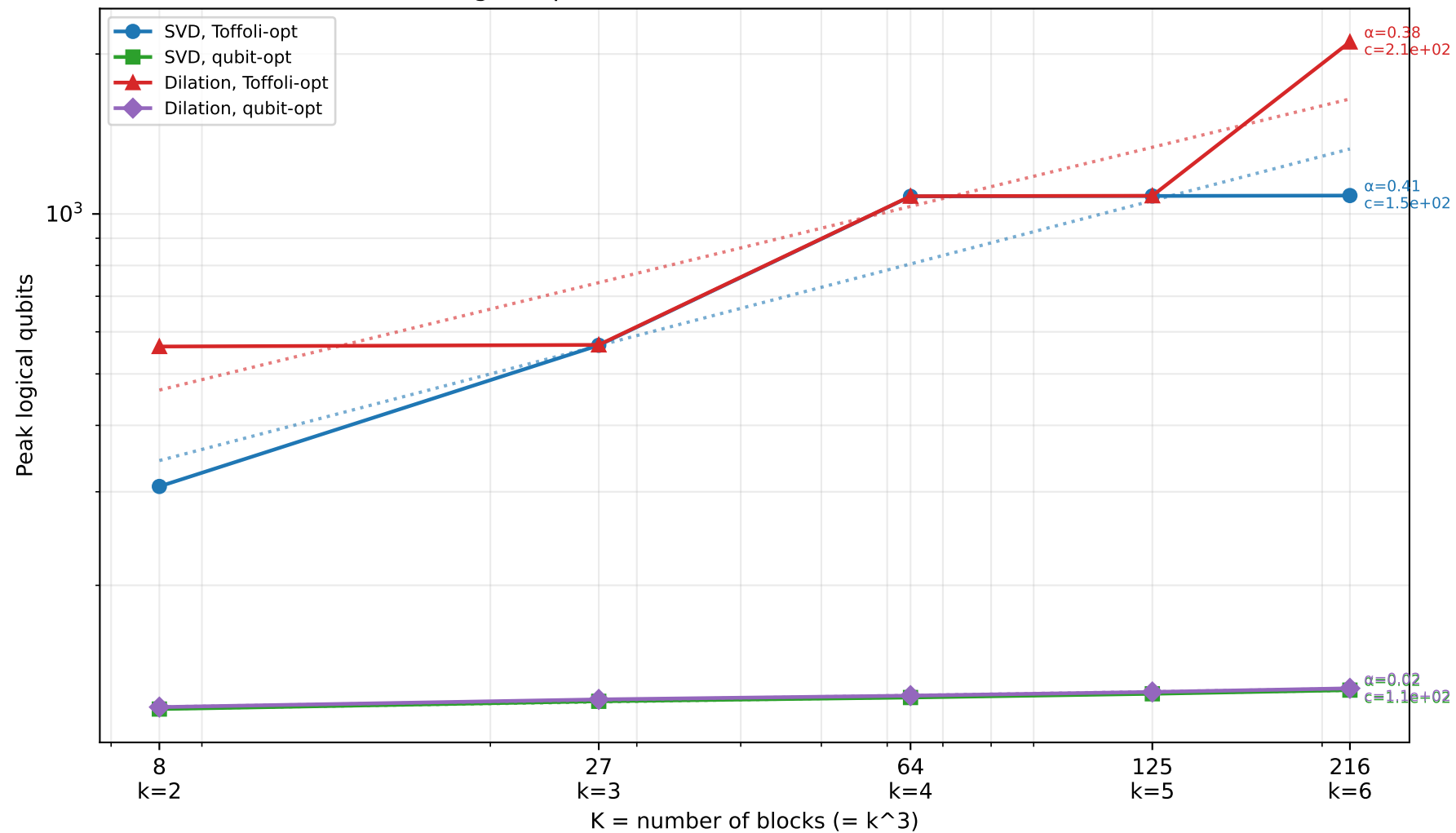
Fits $y = c * K^\alpha$:

SVD, Toffoli-opt	T: $\alpha=0.482$, $c=1.081e+05$; Q: $\alpha=0.410$, $c=1.465e+02$
SVD, qubit-opt	T: $\alpha=0.968$, $c=7.860e+04$; Q: $\alpha=0.024$, $c=1.114e+02$
Dilation, Toffoli-opt	T: $\alpha=0.470$, $c=1.585e+05$; Q: $\alpha=0.383$, $c=2.103e+02$
Dilation, qubit-opt	T: $\alpha=0.983$, $c=1.438e+05$; Q: $\alpha=0.024$, $c=1.124e+02$

Toffoli count: SVD vs direct dilation (N=256, b=32)



Peak logical qubits: SVD vs direct dilation (N=256, b=32)



Resource counts (fixed N=256, b=32)

k	K=k ³	SVD T-opt T	SVD T-opt Q	SVD Q-opt T	SVD Q-opt Q	Dil T-opt T	Dil T-opt Q	Dil Q-opt T	Dil Q-opt Q
2	8	296,302	307	598,728	117	426,240	563	1,121,305	118
3	27	519,666	566	1,868,308	121	739,968	567	3,631,129	122
4	64	824,614	1079	4,340,584	123	1,084,936	1080	8,518,681	124
5	125	1,078,984	1081	8,416,482	125	1,588,552	1082	16,576,537	126
6	216	1,458,366	1083	14,496,832	127	1,973,016	2107	28,597,273	128